25900558 **Reg.No : .....**

Name : .....

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) Regular EXAMINATION October 2025

Third SEMESTER

Discipline Specific Elective (DSE) - MG3DSEPHY200 - BASIC SEMICONDUCTOR PHYSICS

(2024 ADMISSION ONWARDS)

Duration: 1 Hour 30 Minutes **Maximum Marks: 50**

Remember(K), Understand(U), Apply(A), Analyse(An), Evaluate(E), Create(C), Skill(S), Interest(I) and Appreciation(Ap)

Students should attempt at least one question from each course outcome to enhance their overall outcome attainability.

Part A

Short Answer Type Questions

Answer any **10** questions

Each question carries **2** marks

1. Write the relation of transition capacitance with reverse bias voltage . [K] / [CO1]
2. Write the values of barrier potential of Silicon PN Junction and Germanium PN Junction at room temperature. [K] / [CO1]
3. List any two applications of a zener diode. [K] / [CO2]
4. What is dynamic resistance of a diode? [K] / [CO2]
5. Draw the V-I characteristics curve for a PN junction diode. [K] / [CO2]
6. A full-wave rectifier gives a DC output of 50 V and an RMS value of ripple 21 V. Find the ripple factor. [A] / [CO3]
7. Why is half-wave rectifier not preferred for power supply applications? [U] / [CO3]
8. What is the difference between Peak Inverse Voltage of a centre tapped and bridge rectifier? [U] / [CO3]
9. Why is it important to choose the Q-point at the middle of the load line? [E] / [CO4]
10. Define α and β in transistor configuration . [U] / [CO4]

11. Draw the block diagram of voltage-shunt feedback [U] / [CO5]

12. Define current gain of a CE amplifier [U] / [CO5]

13. Define bandwidth and gain-bandwidth product of an amplifier. [U] / [CO5]

14. Name the essential components of a feedback LC oscillator. [K] / [CO6]

[2x10 = 20]

Part B

Short Essay Type Questions

Answer any **6** questions

Each question carries **5** marks

15. What happens when a PN junction is forward biased? Draw a forward biased PN junction [K] / [CO1]

16. With neat circuit diagram explain the working of a Zener diode voltage regulator. [K] / [CO2]

17. Draw the circuit diagram of an inductor filter and explain its working with waveform. [U] / [CO3]

18. Explain the functions of emitter, base and collector regions of a transistor. [U] / [CO4]

19. Draw transfer characteristics of a transistor in CE configuration. What is its importance? [An] / [CO4]

20. With the help of a diagram, explain the working principle of a feedback amplifier. [U] / [CO5]

21. In a CE transistor amplifier with voltage divider bias, show that the resistors R_1 and R_2 together with the emitter resistor could provide proper biasing and stability to the circuit [U] / [CO5]

22. With necessary circuit diagram explain the working of Colpitts oscillator. [U] / [CO6]

[5x6 = 30]